

## NOVI LIGURE, Italy

WMO: 161180

Lat: 44.767N

Lon: 8.783E

Elev: 187

StdP: 99.10

Time Zone: 1.00 (EUC)

Period: 86-98

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -6.1       | -4.2 | -13.2                         | 1.2 | 6.3  | -9.9 | 1.7 | 5.3  | 8.9                   | 6.4  | 7.2 | 6.0  | 1.3                   | 140  | 0.403 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 11.0                   | 31.4            | 19.9 | 30.2 | 19.9 | 29.1 | 19.2 | 23.2                | 28.6 | 22.4 | 27.4 | 21.6 | 26.3 | 2.2                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 21.2                            | 16.3 | 25.4 | 20.9 | 15.9 | 25.2 | 19.9 | 15.0 | 24.6 | 69.6          | 28.8 | 66.4 | 27.2 | 63.6 | 26.3 | 25.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 6.6           | 5.3  | 4.3 | -7.7                       | 33.1 | 2.4                | 0.9 | -9.4                                               | 33.7 | -10.8      | 34.3 | -12.1      | 34.8 | -13.8      | 35.4 |    |    |
| (5)               |      |     | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |         | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|---------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |         | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg   | 12.8        | 2.0   | 4.7   | 8.9   | 11.6  | 17.3  | 20.1  | 23.1  | 23.4  | 19.3  | 13.6  | 6.9   | 2.7   |  |  |
|                                                                        | DBStd   | 8.05        | 2.76  | 3.89  | 3.38  | 2.70  | 3.20  | 2.75  | 2.35  | 2.35  | 2.49  | 3.42  | 2.90  | 3.16  |  |  |
|                                                                        | HDD10.0 | 811         | 247   | 151   | 61    | 14    | 1     | 0     | 0     | 0     | 0     | 11    | 99    | 228   |  |  |
|                                                                        | HDD18.3 | 2450        | 505   | 381   | 293   | 201   | 57    | 14    | 1     | 1     | 18    | 149   | 344   | 486   |  |  |
|                                                                        | CDD10.0 | 1847        | 0     | 3     | 26    | 63    | 226   | 302   | 406   | 417   | 279   | 122   | 4     | 0     |  |  |
|                                                                        | CDD18.3 | 445         | 0     | 0     | 0     | 0     | 23    | 66    | 148   | 159   | 47    | 2     | 0     | 0     |  |  |
|                                                                        | CDH23.3 | 3469        | 0     | 0     | 1     | 1     | 156   | 439   | 1266  | 1318  | 275   | 14    | 0     | 0     |  |  |
|                                                                        | CDH26.7 | 1000        | 0     | 0     | 0     | 0     | 19    | 87    | 407   | 449   | 38    | 1     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg   | 1.4         | 1.0   | 1.3   | 1.7   | 1.9   | 1.8   | 1.8   | 1.6   | 1.5   | 1.3   | 1.1   | 0.9   | 1.0   |  |  |
| (15) Precipitation                                                     | PrecAvg | 838         | 66    | 58    | 47    | 72    | 57    | 50    | 31    | 55    | 98    | 116   | 126   | 65    |  |  |
|                                                                        | PrecMax | 1328        | 273   | 152   | 157   | 182   | 162   | 126   | 82    | 121   | 289   | 262   | 420   | 177   |  |  |
|                                                                        | PrecMin | 515         | 2     | 3     | 9     | 5     | 0     | 1     | 2     | 2     | 4     | 26    | 4     | 0     |  |  |
|                                                                        | PrecStd | 230         | 65    | 46    | 38    | 52    | 36    | 36    | 21    | 36    | 79    | 67    | 103   | 49    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%    | DB          | 11.8  | 17.7  | 22.1  | 22.8  | 28.7  | 30.9  | 32.9  | 33.0  | 29.7  | 25.1  | 15.2  | 12.8  |  |  |
|                                                                        |         | MCWB        | 6.4   | 10.8  | 12.9  | 14.9  | 19.0  | 21.0  | 20.4  | 17.1  | 18.9  | 15.2  | 10.7  | 7.6   |  |  |
|                                                                        | 2%      | DB          | 8.9   | 14.8  | 19.1  | 20.7  | 26.9  | 29.2  | 31.5  | 31.7  | 27.8  | 22.1  | 13.5  | 9.8   |  |  |
|                                                                        |         | MCWB        | 6.2   | 9.6   | 11.2  | 13.1  | 17.8  | 19.6  | 20.4  | 18.9  | 19.1  | 15.5  | 10.4  | 7.3   |  |  |
|                                                                        | 5%      | DB          | 7.2   | 12.7  | 17.1  | 18.9  | 25.0  | 27.8  | 30.2  | 30.7  | 26.2  | 20.0  | 12.2  | 8.1   |  |  |
|                                                                        |         | MCWB        | 5.6   | 8.3   | 10.4  | 12.3  | 16.3  | 18.4  | 20.2  | 19.6  | 17.9  | 14.3  | 9.7   | 6.5   |  |  |
|                                                                        | 10%     | DB          | 6.1   | 10.9  | 15.1  | 17.1  | 23.4  | 26.0  | 29.1  | 29.3  | 24.9  | 18.1  | 11.1  | 7.1   |  |  |
|                                                                        |         | MCWB        | 4.9   | 7.8   | 9.7   | 11.4  | 15.6  | 17.5  | 19.3  | 19.1  | 17.2  | 14.0  | 9.2   | 5.6   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%    | WB          | 8.4   | 11.9  | 13.9  | 15.5  | 21.0  | 23.6  | 24.3  | 24.1  | 22.8  | 18.0  | 12.8  | 10.5  |  |  |
|                                                                        |         | MCDB        | 9.5   | 18.0  | 20.3  | 20.8  | 27.8  | 29.5  | 29.3  | 29.7  | 27.0  | 21.8  | 14.5  | 11.5  |  |  |
|                                                                        | 2%      | WB          | 7.6   | 10.3  | 12.5  | 14.1  | 19.6  | 21.6  | 23.1  | 23.2  | 21.4  | 17.1  | 11.9  | 8.6   |  |  |
|                                                                        |         | MCDB        | 8.3   | 14.1  | 18.4  | 18.5  | 24.9  | 26.9  | 28.4  | 28.7  | 25.4  | 20.0  | 13.3  | 9.6   |  |  |
|                                                                        | 5%      | WB          | 6.6   | 9.7   | 11.5  | 13.1  | 18.0  | 20.4  | 22.4  | 22.4  | 20.1  | 16.3  | 10.6  | 7.7   |  |  |
|                                                                        |         | MCDB        | 7.6   | 12.6  | 16.7  | 16.6  | 22.8  | 25.3  | 27.5  | 27.3  | 23.8  | 18.7  | 12.0  | 8.5   |  |  |
|                                                                        | 10%     | WB          | 5.6   | 8.9   | 10.7  | 12.3  | 16.9  | 19.4  | 21.5  | 21.6  | 19.2  | 15.5  | 10.1  | 6.6   |  |  |
|                                                                        |         | MCDB        | 6.5   | 11.1  | 14.5  | 16.1  | 21.2  | 24.0  | 26.5  | 26.0  | 22.2  | 17.4  | 11.1  | 7.4   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB   | MDBR        | 6.0   | 8.4   | 10.4  | 8.9   | 10.0  | 9.8   | 10.9  | 11.0  | 10.0  | 7.3   | 6.2   | 6.1   |  |  |
|                                                                        |         | MCDBR       | 8.6   | 11.9  | 13.6  | 12.3  | 12.3  | 11.5  | 12.2  | 12.3  | 11.6  | 10.5  | 8.7   | 8.4   |  |  |
|                                                                        |         | MCWBR       | 5.8   | 6.6   | 6.6   | 5.9   | 4.5   | 3.9   | 4.1   | 3.9   | 4.1   | 4.4   | 4.9   | 6.0   |  |  |
|                                                                        |         | MCWBR       | 6.4   | 11.5  | 12.7  | 9.1   | 10.3  | 9.8   | 10.2  | 10.1  | 9.5   | 7.6   | 4.6   | 5.4   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub    | taub        | 0.336 | 0.359 | 0.403 | 0.444 | 0.443 | 0.462 | 0.448 | 0.441 | 0.430 | 0.408 | 0.364 | 0.332 |  |  |
|                                                                        |         | taud        | 2.415 | 2.330 | 2.217 | 2.121 | 2.175 | 2.167 | 2.212 | 2.241 | 2.239 | 2.269 | 2.360 | 2.443 |  |  |
|                                                                        |         | Ebn at Noon | 771   | 815   | 822   | 817   | 831   | 814   | 820   | 812   | 786   | 744   | 722   | 737   |  |  |
|                                                                        |         | Edh at Noon | 75    | 97    | 123   | 148   | 144   | 146   | 138   | 130   | 120   | 102   | 79    | 67    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg  | RadAvg      | 1.50  | 2.27  | 3.59  | 4.63  | 5.74  | 6.61  | 6.80  | 5.78  | 4.32  | 2.53  | 1.52  | 1.26  |  |  |
|                                                                        |         | RadStd      | 0.20  | 0.34  | 0.47  | 0.46  | 0.47  | 0.35  | 0.37  | 0.34  | 0.28  | 0.37  | 0.23  | 0.18  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)     | (m)     |         |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |
| (47) Regional (43 neighbors) |  | +0.36 | N/A     | N/A    | +0.60 | N/A     | N/A   | N/A | N/A         | +77     | +23     |         |

Nomenclature: See separate page